

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ 4**DURATION: 25 MINUTES****SUMMER SEMESTER, 2023-2024****FULL MARKS: 20**

CSE 4411: Data Communication and Networking

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

STUDENT ID:

1. In CDMA, all users share the same frequency, but each user has its own code to encode data. Let's consider two users (A, B) where the code for A, $C_A = 1, 1, 1, -1, 1, -1, -1, -1$, and the code for B, $C_B = 1, -1, 1, 1, 1, -1, 1, 1$. Now answer the following questions based on the given information. 5 + 1 +
2 + 2
(CO3)
(PO2)

- a) Explain the CDMA process using the given values. You can take any data for demonstration. (Only show for a single time slot)

Solution:

Same as slides

- b) If we flip each of the values of C_B , can we still send data with minimal interference? Why? Why not?

Solution:

Yes, because they are still orthogonal: $C_A \cdot C'_B = C_A(-C_B) = -(C_A \cdot C_B)$

- c) Provide two new codes for A and B where at least 4 values are changed, but will still provide with minimum interference

Solution:

$C_A = -1, 1, -1, 1, 1, 1, 1, 1$ $C_B = -1, -1, -1, -1, 1, -1, 1, -1$

2. Suppose the router between subnet 2 and subnet 3 is called router R1, and the router between subnet 1 and subnet 2 is called router R2 4 +
4 + 2
(CO3)
(PO2)

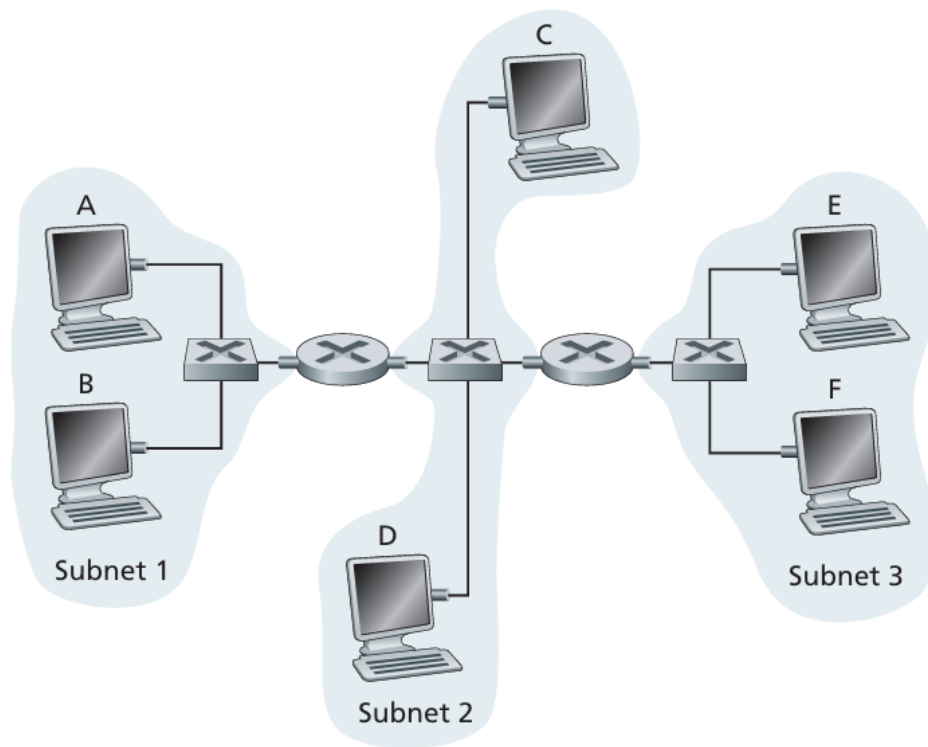


Figure 1: Topology for Question 2

- a) Consider sending an IP datagram from Host E to Host F. Will Host E ask router R1 to help forward the datagram? Why? In the Ethernet frame containing the IP datagram, what are the source and destination IP and MAC addresses?

Solution:

No. E can check the subnet prefix of Host F's IP address, and then learn that F is on the same LAN. Thus, E will not send the packet to the default router R1.

Ethernet frame from E to F:

Source IP = E's IP address

Destination IP = F's IP address

Source MAC = E's MAC address

Destination MAC = F's MAC address

- b) Suppose E would like to send an IP datagram to B, and assume that E's ARP cache does not contain B's MAC address. Will E perform an ARP query to find B's MAC address? Why? In the Ethernet frame (containing the IP datagram destined to B) that is delivered to router R1, what are the source and destination IP and MAC addresses?

Solution:

No, because they are not on the same LAN. E can find this out by checking B's IP address.

Ethernet frame from E to R1:

Source IP = E's IP address

Destination IP = B's IP address

Source MAC = E's MAC address

Destination MAC = The MAC address of R1's interface connecting to Subnet 3.

- c) Is it possible to put E and F into different subnets without changing the topology only changing the configuration? How?

Solution:

Yes, using VLAN